OUTCOMES-BASED EDUCATION (OBE) COURSE SYLLABUS IN
CC103 – Computer Programming 2
Date: January 2026

• **School Year/ Semester**	AY 2025-2026 / 2nd Semester
• **University**	Vision: A globally-recognized green research University committed to excellent service to the nation. Mission: Foster excellence, holistic and inclusive education attuned to the demands of a diverse world towards the formation of productive and value-laden lives Goal: Uphold the tradition of excellence in instruction, research, extension ad production geared towards a strong academe-industry interface for societal development Core Values: Respect, Integrity, Social Responsibility, Excellence, Commitment Institutional Outcomes: • A *Professiona*l who is morally upright, socially responsible and globally employable • A *Leader and Innovator* who inspires others and is committed to social transformation and national development • An *Environmental Advocate* committed to research, extension and production initiatives
• **College/Campus**	College of Information and Communications Technology/ Main Campus
• **Program/Degree**	Bachelor of Science in Information Technology (BSINFOTECH)
• **Program Outcomes**	

• Articulate and discuss the latest developments in the specific field of practice.	**I**
• Effectively communicate orally and in writing using both English and Filipino.	**P**
• Work effectively and independently in multi-disciplinary and multi-cultural teams.	**P**
• Act in recognition of professional, social and ethical responsibility.	**I**
• Preserve and promote “Filipino historical and cultural heritage”.	**I**
• Apply knowledge of computing science, and mathematics appropriate to the discipline.	**D**
• Understand best practices and standards and their applications.	**P**
• Analyze complex problems and identify and define the computing requirements appropriate to its solution.	**D**
• Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.	**I**
• Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.	**D**
• Integrate IT-based solutions into the user environment effectively.	**P**
• Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.	**D**
• Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.	**P**
• Assist in the creation of an effective IT project plan.	**I**

• Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.	**P**
• Analyze the local and global impact of computing information technology on individuals, organizations, and society.	**I**
• Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.	**I**
• Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.	**P**
• Participate in the generation of new knowledge or in research and development projects	**I**
• Support local, regional and national development plans along education, environment, socio economic, health, gender and development, science and technology.	**I**

I – Introduced **P** - Practiced **D** - Demonstrated

- **Course Code/Title**: CC103 – Computer Programming 2
- **Course Description**: This course is a continuation of CC102 – Computer Programming 1. The emphasis is to train students to design, implement, test, and debug programs intended to solve computing problems using basic data structures and standard libraries.
- **Course Credit**: 3 units (2 units lecture, 1 unit laboratory)
- **Prerequisite**: CC102 – Computer Programming 1
- **Contact Hours**: 5 hours / week (2 hours lecture, 3 hours laboratory) (90 hours per semester)
- **Course Outcomes**:

At the end of the course, the students shall be able to:

COs	Description	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
CO1	Design, implement, test, and debug modular programs using functions, enabling efficient and reusable code to solve computational problems.						3	3					3	1		1			2		
CO2	Apply the concepts of pointers and dynamic variables to manage memory effectively, ensuring flexibility and efficiency in program execution.						3	2	3		2		3					1			
CO3	Demonstrate proficiency in using arrays, structures, and linked lists to represent and manipulate data, enabling robust problem-solving techniques.						3	3			3	2	3				1		2		
CO4	Design programs that perform file input/output operations to store, retrieve, and process data, ensuring long-term persistence of information.						3	2			3	2	3			1					

1 – CO has minor contribution to PO **2** – CO has moderate contribution to the PO **3** – CO has major contribution to the PO

- **Course Outline/Learning Plan**

POs	COs	Desired Learning Outcomes	Course Content/ Subject Matter	Textbooks/ References	Teaching and Learning Activities *(TLAs)*	Assessment Tasks	Instructional Resources/ Materials	Time Table (hours)
F G L M O R	**CO1**	At the end of the lesson, students shall be able to: • Explain the role of functions in organizing and simplifying code;	• **Functions** • Functions as Subprograms • Return Values and Parameters • Local Variables • Recursive Functions	King, K. N. (2020). C Programming: A Modern Approach (2nd Edition). W.W. Norton & Company. Harbison, S., & Steele, G. L. (2021). C: A Reference	• Lecture • Brainstorming • Interactive discussion • Programming Demonstration • Guided Hands-on Programming Sessions • Online learning through	• Quizzes • Assignment • Activities	• Handouts/ Presentation • Computer/ Laptop/ Cellphones • Internet Connectivity	**10**

		• Create and implement C functions to solve specific tasks in a program; • Describe the use of parameters to pass data between function; and • Distinguish between local and global variables, understanding their scope and usage.	• Function Prototypes • Standard C Function •	Manual (5th Edition). Pearson.	Facebook GC and/or Google Classroom • Blended/Hybrid Learning			
F G H J L Q	**CO2**	At the end of the lesson, students shall be able to: • Define what a pointer is and explain its role in referencing memory locations; • Apply pointer operators to manipulate and access data in memory within a program; and • Pass pointers as function parameters to modify data efficiently within functions.	• **Pointers** • Memory Addresses • Pointer Variables and Operators • Pointers as Parameters	King, K. N. (2020). C Programming: A Modern Approach (2nd Edition). W.W. Norton & Company. Harbison, S., & Steele, G. L. (2021). C: A Reference Manual (5th Edition). Pearson.	• Lecture • Brainstorming • Interactive discussion • Programming Demonstration • Guided Hands-on Programming Sessions • Online learning through Facebook GC and/or Google Classroom • Blended/Hybrid Learning	• Quizzes • Assignment • Activities	• Handouts/ Presentation • Computer/ Laptop/ Cellphones • Internet Connectivity	**15**
F G J K L P R	**C03**	At the end of the lesson, students shall be able to: • Distinguish between structured and simple data types, understanding their differences and applications; • Define an array and understand its purpose in storing multiple values of the same type; and • Implement arrays in a program to store and manipulate data efficiently	• **Arrays** • Structured Data Types • Defining an Array • Array Initialization • Operations on Arrays • Array of Arrays • Pointers and Arrays • Arrays as Function Parameters • Strings	King, K. N. (2020). C Programming: A Modern Approach (2nd Edition). W.W. Norton & Company. Deitel, P. J., & Deitel, H. M. (2020). *C How to Program (8th Edition)*. Pearson.	• Lecture • Brainstorming • Interactive discussion • Programming Demonstration • Guided Hands-on Programming Sessions • Online learning through Facebook GC and/or Google Classroom • Blended/Hybrid Learning	• Quizzes • Assignment • Activities	• Handouts/ Presentation • Computer/ Laptop/ Cellphones • Internet Connectivity	**17.5**

MIDTERM EXAMINATION	**2.5**							
F G J K L P R	**CO3**	At the end of the lesson, students shall be able to: • Explain the difference between a structure and an array, highlighting their unique features and use; • Define and implement a structure in a program to group related data; and • Store and retrieve values from a structure variable to manipulate structured data.	• **Structures** • Defining a Structure • Structures as Function Parameters • Array of Structures • Type Definitions	King, K. N. (2020). C Programming: A Modern Approach (2nd Edition). W.W. Norton & Company. Harbison, S., & Steele, G. L. (2021). C: A Reference Manual (5th Edition). Pearson.	• Lecture • Brainstorming • Interactive discussion • Programming Demonstration • Guided Hands-on Programming Sessions • Online learning through Facebook GC and/or Google Classroom • Blended/Hybrid Learning	• Quizzes • Assignment • Activities	• Handouts/ Presentation • Computer/ Laptop/ Cellphones • Internet Connectivity	**15**
F G H J K L P Q R	**CO2 CO3**	At the end of the lesson, students shall be able to: • Describe the basics of how main memory is managed, including allocation and deallocation of memory; • Create and manage dynamic variables within a program, utilizing memory allocation and deallocation techniques; and • Implement and manipulate linked lists to manage collections of data in a dynamic, flexible way.	• **Dynamic Variables and Linked Lists** • Main Memory Management • Creating Dynamic Variables • Self-referential Structures • Linked Lists • Operations on Linked Lists	King, K. N. (2020). C Programming: A Modern Approach (2nd Edition). W.W. Norton & Company. Deitel, P. J., & Deitel, H. M. (2020). *C How to Program (8th Edition)*. Pearson.	• Lecture • Brainstorming • Interactive discussion • Programming Demonstration • Guided Hands-on Programming Sessions • Online learning through Facebook GC and/or Google Classroom • Blended/Hybrid Learning	• Quizzes • Assignment • Activities	• Handouts/ Presentation • Computer/ Laptop/ Cellphones • Internet Connectivity	**17.5**
F G J K L O	**CO4**	At the end of the lesson, students shall be able to: • Distinguish between text files and binary files, understanding their structure and usage;	• **Files** • Bytes and Characters • Files and File Types • File Pointers, Fopen and Fclose • Reading and Writing Bytes	King, K. N. (2020). C Programming: A Modern Approach (2nd Edition). W.W. Norton & Company. Harbison, S., & Steele, G. L. (2021). C: A Reference Manual (5th	• Lecture • Brainstorming • Interactive discussion • Programming Demonstration • Guided Hands-on Programming Sessions • Online learning through	• Quizzes • Assignment • Activities	• Handouts/ Presentation • Computer/ Laptop/ Cellphones • Internet Connectivity	**10**

		• Define, open, and close a file in a program to manage file operations; • Use file handling statements to store data into a file; and • Retrieve data from a file using file handling statements to read stored values.	• String Input and Output • Formatted Input and Output • Input and Output of Binary Data	Edition). Pearson.	Facebook GC and/or Google Classroom • Blended/Hybrid Learning			
FINAL EXAMINATION	**2.5**							

• **Suggested Readings and References**	• "The C Programming Language" by Brian W. Kernighan and Dennis M. Ritchie (2022 Edition) • "C Programming: A Modern Approach" by K. N. King (2nd Edition, 2022) https://www.amazon.com/C-Programming-Modern-Approach-2nd/dp/0393979504 • Algorithmic Thinking, 2nd Edition Learn Algorithms to Level Up Your Coding Skills by Daniel Zingaro. Retrieved January 7, 2025, from https://nostarch.com/algorithmic-thinking-2nd-edition • C Tutorial. Retrieved January 7, 2025, from https://www.w3schools.com/c/ • What is C Language: Introduction to C Programming By Ravikiran A S. Retrieved January 7, 2025, from https://www.simplilearn.com/tutorials/c-tutorial/use-of-c-language • The C Programming Handbook for Beginners by Dionysia Lemonaki. Retrieved January 7, 2025, from https://www.freecodecamp.org/news/the-c-programming-handbook-for-beginners/ • Learning C Programming. Retrieved January 7, 2025, from https://www.programiz.com/c-programming • C Programming Tutorial. Retrieved January 7, 2025, from https://www.tutorialspoint.com/cprogramming/index.htm
• **Course Requirements**	• Written outputs in this course such as assignments, quizzes, seatwork, and problem set or exercises, must be submitted before the end of the semester or special permission must be requested from the Instructor/Professor before the due date. • Take Midterm and Final examinations at the online or at the campus.
• **Course Policies**	• Students with reasonable absences from the class will be given consideration for the missed quizzes/examinations upon presentation of a duly signed excuse letter. • Students who incurred more than 20% of the required number of hours shall be dropped from the rolls. • Submission of projects/activities shall be on or before the due date agreed upon by the students and the faculty. Late submission will earn a grade of 3.0. • Cheating during examination is strictly prohibited; students caught doing such act shall be dealt with as provided in the Student Handbook. • Cellphone use during classes that is not related to the subject is strictly prohibited. • Wearing of appropriate school uniform and ID shall be implemented. • In the event of class disruptions, the guidelines for Flexible Learning Modality will take effect with the university's approved provisions in shifting classes.
XVI. Grading System (Per Board Resolution No. 3, 4 & 5, S. 2016)	The final grade of each student will be computed as follows: Computation of Scores Written Output 30% Performance 30% Midterm/Final Examination 40% **TOTAL 100%** Distribution of Weight for the Overall/Final Grade Mid-Term Grade 50% Tentative 50%
XVII. Consultation Time/Venue	WED 8:00 – 12:00 | CICT Faculty Office

Prepared by:	Reviewed by:	Approved:
MARY ROSE B. TABLO	**ASTER VIVIEN C. VARGAS, MSIT**	**MARIA CONCEPCION S. VERA, DIT**
Instructor I	Department Chairperson, Computing Programs	Dean, CICT
Date:_____	Date:_____	Date:_____